

Closure aur Variable Scope

Block scope (let/const)

{ } ke andar `let` ya `const` se banaya variable sirf usi block me dikhta hai. `if`, `for`, `while` ke andar bhi yahi rule hai.

```
javascript
```

```
if (true) {  
  let name = "Raj";  
}  
console.log(name); // Error: name defined nahi hai
```

Nested function

Function ke andar function banana nested function kehlata hai. Andar wala function bahar ke (outer) variables use kar sakta hai.

Lexical Environment (simple bhasha me)

- Har function, block aur poori script ke paas ek hidden "dabba" hota hai jisme uske variables rakhe jaate hain.
- Is dabbe ke paas ek **link** bhi hota hai apne bahar wale dabbe ka.
- Variable dhoondhte waqt JS pehle apna dabba dekhta hai, phir bahar wala, phir usse bahar wala... global tak.

Closure kya hai?

Closure = aisa function jo yaad rakhta hai ki wo kahan bana tha, aur wahan ke variables use kar sakta hai, chahe usko kahin se bhi call karo.

JS me saare functions automatically closure hote hain (sirf `new Function` exception hai).

```
javascript
```

```
function makeCounter() {  
  let count = 0;  
  return function () {  
    return count++;  
  };  
}  
  
let c1 = makeCounter();  
let c2 = makeCounter();  
  
c1(); // 0  
c1(); // 1  
c2(); // 0 -> c2 ka apna alag count hai
```

Har `makeCounter()` call par naya dabba banta hai, isliye counters ek dusre se independent hain.

Zaroori baatein

- Function ko variable ki **latest value** milti hai, purani copy nahi.
- `let` variable declare hone se pehle use karoge to error aata hai. Is beech ke time ko "dead zone" kehte hain.
- **Garbage collection:** jab tak koi inner function zinda hai, uska outer dabba memory me rehta hai. Function `null` kar do to memory free ho jaati hai.
- `for (let i...)` me har iteration ka apna alag `i` hota hai. Isliye loop me functions banate waqt `let` ke saath `for` safe rehta hai.

Yaad rakho

1. Function apne birth-place ke variables yaad rakhta hai (closure).
2. Variable wahi update hota hai jahan wo rehta hai.
3. Alag-alag calls ke alag-alag dabbe hote hain.